

$a = 1.96 \text{ m s}^{-2}$
 Using equation (2), $T = 10(1.96) + 10(9.81) = 118 \text{ N}$

4(a) *(This table isn't required, just for your revision.)*

	Progressive Waves	Stationary Waves
Amplitude	Amplitude is the same for all the particles in the path of the wave	Amplitude varies according to positions: from zero at the nodes to a maximum at the antinodes
Frequency	All particles oscillate at the same frequency	All particles oscillate at the same frequency except at the nodes
Wavelength	The wavelength is the distance between any 2 consecutive points on the wave with the same phase	The wavelength is 2 times the distance between a pair of adjacent nodes or antinodes
Phase	All particles within one wavelength have different phases	All particles within the same segment are in phase ; particles in alternate segments are out of phase by 180°
Waveform	Propagates with the speed of the wave	No propagation of the wave profile
Energy	Energy is transmitted in the direction of the travel of the wave	No transmission of energy – energy is retained within vibratory motion of wave

Similarity:

All particles (except the nodes of stationary waves) oscillate at the same frequency.

Difference:

In progressive waves, all the particles have the same amplitude.

In stationary waves, the amplitude of a particle varies according to its position.

- (b) When a stationary wave is formed, nodes must be located at the vibrator and at the top of the pulley wheel.

So the length L of the horizontal string must be divided into integral multiples of half-wavelength ($\lambda/2$). Consequently, the wavelength λ can only take specific values.

OR

Consequently, the wavelength λ can only take specific values: $\lambda = 2L/n$, where $n = 1, 2, 3, \dots$ and L is the length of the horizontal string.

Since the wave speed v is constant and the frequency of vibration is given by $f = v/\lambda$, stationary waves can only form at specific frequencies.

- (c) From the figure, $3\lambda = 75 \text{ cm}$

So, $\lambda = 25 \text{ cm}$

Speed of wave propagation, $v = f\lambda = (120)(0.25) = \mathbf{30 \text{ m s}^{-1}}$